Domain-Specific Fine-Tuning of LLMs for Multilingual Student Advisory in Higher Education

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***Abstract*— Colleges are experiencing an increased need to have smart, multi-lingual support systems that can respond to various questions posed by the students. Although large language models (LLMs) have strong general capabilities, they are often inaccurate, culturally insensitive, and unsafe in low-resource and domain-specific settings. The paper introduces a two-stage fine-tuning model for developing a multilingual student services assistant specializing in English, Modern Standard Arabic (MSA), Franco Arabic, and Omani Arabic. Its purpose is to improve the factual accuracy, dialogue coherence, and refusal accuracy in the sensitive institutional environment. In Phase I, factual memorization was performed by fine-tuning the TyDi QA and MASSIVE datasets with LoRA adapters to enhance multilingual understanding. Phase II used behavioral alignment, including institutional data and the presence of refusal-inducing prompts, to improve the safe management of out-of-scope queries. The evaluation metrics were factual accuracy, BLEU, COMET, refusal precision, and multi-turn coherence, with performance evaluated against GPT-4, Aya-Base, and Vicuna. The suggested model was more successful in refusal accuracy (0.86 vs. GPT-4 at 0.82) and competitive accuracy (0.78 vs. GPT-4 at 0.81). It is noteworthy that the dialectal performance was enhanced, especially in the Omani Arabic. Domain-specific finetuning can be used to create safer and more inclusive multilingual LLM assistants in higher education and find the balance between accuracy and reliability between underrepresented languages.**

Keywords—Large Language Models, Fine-Tuning, TyDi QA, MASSIVE, Multilingual, Higher Education, Student Support

I. INTRODUCTION

Advancements in multilingual comprehension and production have been stunning with Large Language Models (LLMs), which can be used in education, health care, and customer support [1, 2]. Their ability to handle multiple languages makes them a powerful tool for addressing communication challenges in higher education institutions that are becoming increasingly multilingual. Cross-lingual evaluation and adaptation have also been improved using benchmarks like TyDi QA [3], MASSIVE [4], and newer developments in parameter-efficient fine-tuning techniques, including Low-Rank Adaptation (LoRA) have made adapting foundational models to specific fields less computationally intense [5]. Recent advances in educational AI and the growing use of LLM-based systems in academic institutions support the topicality of the work. This research can also be considered part of the current trends in responsible and inclusive AI implementation, as it addresses the needs of multilingual and dialect-sensitive students and helps fill gaps in the development of LLMs in higher education. Although these innovations have been achieved, there is a shortage of such multilingual assistants in practical use in universities, especially for regional dialects and domain-specific queries. Internationalization is driving linguistic diversity in universities around the globe and variations in language across regions [6]. Students often ask questions on admissions, course requirements, examinations, and institutional policies in various languages and dialects, including English, Modern Standard Arabic (MSA), Franco-Arabic, and Omani Arabic. Traditional rule-based chatbots do not tend to recognize codeswitching, dialectal variation, or context-dependent knowledge [7, 8]. Despite their power, pre-trained LLMs are prone to hallucinating or responding inappropriately to queries in specific domains or unrepresented dialects [9]. These limitations highlight the importance of fine-tuned multilingual systems that can provide accurate, inclusive, and context-sensitive answers. Our methodology is flexible across linguistically diverse educational settings, incorporating multilingual and dialectal variations in English, Modern Standard Arabic, Franco Arabic, and Omani Arabic. This inclusivity and multilingualism make the model applicable across all areas rather than just within a region. They are beneficial to global institutions that face the same challenges in communicating with students and in academic advising. Although general-purpose LLMs perform well on open-domain benchmarks, they exhibit various shortcomings when fine-tuned for student services [10]. First, the factual grounding is not always enough, and the mistake occurs in the most important spheres of academic rules. Second, the refusal behavior is inconsistent: models can answer sensitive questions with speculative responses rather than avoid them as they should [11]. Third, there is a lack of multilingual standards that reflect regional dialects, which is also prevalent in the daily communication between students. Therefore, we focus on academic and administrative questions (e.g., admissions, exams, policies) in multilingual universities. In contrast to earlier PEFT techniques, which focus on task adaptation, our approach is the only one to incorporate both the factual grounds and dialect-sensitive refusal behavior, which are crucial in advising students on trustworthiness. The concept of

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student support in this research paper refers explicitly to academic and administrative advisory contacts, including questions about admissions, course registration, examinations, and institutional policies. The limited scope ensures the proposed system directly addresses the fundamental communication requirements of higher education advisory services. Therefore, higher institutions are challenged to implement powerful AI assistants that are not only fair, reliable, and safe but also accessible to students across various groups. This paper contributes the following to deal with these problems:

- Propose a two-phase fine-tuning approach consisting of factual memorization using behavioral alignment with institutional data and refusal-inducing prompts.
- By integrating TyDi QA and MASSIVE with an institutional corpus, we leverage complementary strengths, cross-lingual comprehension, intent recognition, and domain-specific knowledge.
- Conduct a comprehensive evaluation of the fine-tuned model against state-of-the-art baselines (GPT-4, AyaBase, and Vicuna), using metrics such as factual accuracy, refusal precision, BLEU, COMET, and multi-turn dialogue coherence.

With these contributions, this work will advance the evolution of inclusive and trustworthy AI assistants in higher education by closing the gap between large-scale multilingual benchmarks and domain-specific institutional requirements. The remaining part of the paper is organized in the following way: Related Work provides the reviews of multilingual LLMs, fine-tuning, and benchmarks; System Architecture provides the overview of the five-layer framework; Methodology delivers the information about datasets, fine-tuning pipeline, and evaluation setup; Results and Analysis provides the presentation of the empirical findings with baseline comparisons; Discussion shows the strengths, challenges, and improvements; Conclusion and Future Work provides the insights about the contributions and proposals to add to the existing system, including retrieval augmentation, dataset expansion, and multimodal support.

II. Literature Review

There is an increasing number of AI-driven tools being implemented in higher education across tutoring and grading, as well as administration. Surveys have demonstrated that LLMs can offer scalable and customized answers, thus raising the level of student engagement and institutional knowledge accessibility [1, 2]. Indicatively, Xu et al. [3] emphasize the potential of LLMs in the education sector, but they warn about issues such as fairness and inclusiveness in a multilingual setting. Recent literature also emphasizes the significance of refusal actions and reliability because unsuitable responses might weaken the credibility of institutions [4, 5]. Despite these advances, few studies have investigated domain-specific multilingual assistants tailored to university services. Research in multilingual NLP has advanced rapidly with the introduction of models such as mBERT, XLM-R, and multilingual GPT variants. These models demonstrate strong zero-shot transfer capabilities but often struggle with dialectal variations and domain-specific queries [6]. Fine-tuning techniques such as LoRA [7] and other parameter-efficient approaches (PEFT) enable cost-effective adaptation of large models to specialized domains. Gao et al. [8] survey Retrieval-Augmented Generation (RAG) as a complementary method for grounding responses in external knowledge bases, further reducing hallucination risks. Recent studies in multilingual fine-tuning confirm that domain adaptation improves factual accuracy and refusal precision in specialized settings, but challenges remain in ensuring long-context coherence [9][10]. Two benchmark datasets are central to multilingual evaluation. TyDi QA, with 204K question–answer pairs across 11 typologically diverse languages, has become a standard for multilingual question answering [6]. MASSIVE offers over one million annotated utterances in 51 languages, designed for intent classification and slot filling in conversational AI [7]. While both datasets enhance cross-lingual evaluation, they lack representation of dialects such as Franco-Arabic or Omani Arabic, which are widely used in university contexts in the MENA region. To bridge this gap, our study combines TyDi QA and MASSIVE with an institutional dataset covering English, MSA, Franco-Arabic, and Omani Arabic to develop a domain-specific multilingual assistant.

III. System Architecture and Methodology

The entire system will provide multilingual support to students, ensuring accuracy and safety. It is based on a five-layer architecture, with input normalization to support a wide range of languages and scripts, natural language understanding, a domain-adapted LLM core, and response management and system integration. All the layers have a specific role in ensuring that queries are correctly understood, contextually defined, and in line with institutional policies, and that they exhibit reliable refusal behavior when required. Our architecture is depicted in **Fig. 1**. The input layer standardizes queries in any language using language detection and transliteration.

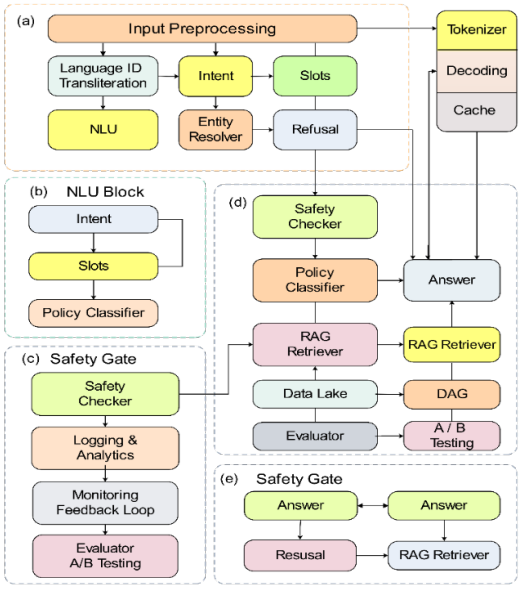


Fig. 1. System architecture diagram

The NLU layer also classifies intents and fills slots using a multilingual transformer fine-tuned on the MASSIVE dataset. The LLM core layer consists of an LLM trained on TyDi QA and an institutional data domain, adapted with

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LoRA-based adapters. The response management layer identifies the intent and content policies to answer or refuse, and the refusal patterns are reinforced during fine-tuning to ensure safety. This last is the integration and logging layer, which interacts with the university systems and logs interactions for continuous improvement.

A. Datasets and Two-Phase Fine-Tuning

We have three sources of training data. TyDi QA [11] offers 204K pairs of queries and answers in 11 languages. MASSIVE [12] provides intent and slot-annotated utterances in more than one million parallel utterances in 51 languages. The institutional dataset consists of 7,000 student queries and official responses in English, MSA, Franco-Arabic, and Omani Arabic. Although our institutional dataset (7K samples) includes the major student services, in the future, coverage will be expanded to provide robustness across both query types and dialects. The fine-tuning process is done in two stages: the first involves memorizing facts, and the second involves aligning behaviors. The initial phase consists of fine-tuning the LLM core using the TyDi QA and MASSIVE datasets with LoRA adapters to enable the model to internalize factual knowledge and improve multilingual understanding. During the second stage, the model undergoes retuning on prompts that trigger refusal and on institutional data to learn how to respond appropriately to questions outside the scope of the institutional setting or lacking sufficient supporting information. This domain-specific fine-tuning plan was created to address the specialized nature of student advisory interactions, where accuracy, institutional conformity, and culturally appropriate refusal behavior are necessary. The model can be improved to align with both factual knowledge and the behavioral pattern of responses to university policies, making it more applicable to real-world academic advisory practice. Dual-process learning theory is the basis of dividing factual memorization and behavioral alignment since it differentiates between knowledge acquisition and context-specific thinking. Phase I and Phase II, respectively, provide the model with multilingual factual competence and behavioral judgment, enabling it to conform to institutional norms and to follow safe response strategies. This hierarchical division provides a factual basis, and ethical refusal behaviors are acquired separately but with a unifying effect, which differs from the theoretical explanations of the selected design and distinguishes the traditional single-phase fine-tuning from the proposed approach.

B. Evaluation Setup

We test the model on out-of-sample parts of the datasets. They are factual accuracy, refusal precision, BLEU, COMET, and coherence. Some of the baselines include GPT-4, Aya-Base, and Vicuna. The correctness and the suitability of refusal are determined by human evaluators who also understand the target languages. Multi-turn coherence is tested on a maximum of five turns in synthetic dialogues. We use three supplementary datasets to support domain-specific fine-tuning, including both general multilingual coverage and institutional context. TyDi QA has 204K pairs of question-answer sentences in 11 typologically diverse languages, whereas MASSIVE has more than a million annotated utterances in 51 languages and 18 domains (see **TABLE 1**). To fill the gap between our benchmark resources and real applications, we also include an institutional dataset of 7,000 student queries and verified answers in four languages, including dialectal Arabic. Collectively, these datasets enable the development of a multilingual assistant that cannot only answer standardized evaluation questions but also more real-world student service questions.

TABLE 1. DATASET STATISTICS (TYDI QA, MASSIVE, INSTITUTIONAL)

Dataset	Languages	Samples	Notes
TyDi QA	11	204K	Q&A pairs, 11 languages
MASSIVE	51	>1M	Utterances, 18 domains
Institutional	4	7K	Student services Q&A

A further detail of sample distribution across the languages in the three datasets in Table 2 shows that the most common languages dominate, but some region-specific varieties are also present. The total share of English is the largest and it is highly represented in all the datasets, most especially MASSIVE, with 500,000 English samples. Another commonality is seen in Modern Standard Arabic (MSA), which is also found in consistent datasets, just like it is essential in multilingual and Arabic-oriented applications. Franco-Arabic and Omani Arabic cannot be found in TyDi QA, but are present in both MASSIVE and the Institutional dataset, providing evidence of the inclusion of these hybrid and dialectal patterns, which are frequently missing in mainstream resources. The Other category, which covers other languages, accounts for a large share of TyDi QA (129,000) and MASSIVE (300,000), but is not reflected in the Institutional data, which is focused and relatively small. Such a distribution shows the balancing forces of the datasets: a wide regional coverage of the languages of the world, data on regional dialects, and domain-specific data. To filter the performance, we employed a multilingual transformer backbone (XLM-R Base) and LoRA adapters (rank = 8, 64, 0.16). The model was trained for 3 epochs; the batch size was 32, the learning rate was $2e^{-5}$, and the AdamW optimizer was used; mixed precision was used. At least 120K steps were executed on four A100 GPUs. Including this information will enhance the reproducibility and transparency of the experimental setup.

TABLE 2. DISTRIBUTION OF SAMPLES BY LANGUAGE AND DATASET

Language	TyDi QA	MASSIVE	Institutional
English	50K	500K	3K
MSA	25K	100K	2K
Franco-Arabic	0	50K	1.2K
Omani Arabic	0	50K	1.8K
Other	129K	300K	0

All the evaluation metrics adopted in this paper are summarized in **TABLE 3**, and all of them describe various aspects of the system performance. Factual Accuracy: It is a measure of the consistency with which facts directly related to the results of getting correct answers are retrieved. In rejecting unanswerable or unsafe queries, which trade informativeness for safety, Refusal Precision is used to test the system. The BLEU score is used as a measure of traditional surface-level similarity between generated and reference responses, based on BLEU-4 n-gram overlap. To this end, the COMET score measures semantic adequacy and fluency, giving greater weight to what is retained in

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terms of meaning and naturalness. Finally, Coherence measures the consistency of single turns, or such that the reaction to earlier turns in dialogue is coherent. A combination of these metrics will provide a comprehensive framework for measuring both the correctness and the quality of the model outputs.

TABLE 3. EVALUATION METRICS USED IN THIS STUDY

Metric	Description
Factual Accuracy	Proportion of correct answers
Refusal Precision	Correct refusals / all refusals
BLEU Score	BLEU-4 n-gram overlap
COMET Score	Semantic adequacy/fluency
Coherence	Consistency across dialogue turns

We determine factual accuracy by measuring the perfect coincidence between generated and ground-truth answers. The ratio of correctly refused queries to the total number of refusals by the query is known as the refusal precision. BLEU and COMET n-gram overlap measures and semantic adequacy measures, respectively, were generated for reference answers. Coherence also assesses the uniformity of multi-turn conversations; here, we measure pairwise semantic similarity across turns. The evaluation has been conducted individually on each language and dataset.

IV. Results and Analysis

This section presents the empirical results of our domain-specific fine-tuning strategy, tested on TyDi QA, MASSIVE, and institutional data. Our findings include data on factual accuracy, refusal accuracy, multilingual response fluency (BLEU and COMET), and dialogue coherence in multi-turn dialogues. It also has comparative analyses to state-of-the-art baselines such as GPT-4, Aya-Base, and Vicuna. The outcomes show the advantages of the proposed pipeline and its limitations in meeting the needs of multilingual students. **TABLE 4** shows the results of factual accuracy using datasets and the languages, and the differences in model performance as adjusting both the type of resource and the language. In the context of TyDi QA, the accuracy is relatively even, with English scoring 0.67 and MSA slightly lower at 0.63. In the case of MASSIVE, the English lag is the lowest at 0.60, and MSA is better at 0.64. The dialectal varieties are significantly higher in performance, with 0.72 and 0.76 on Franco-Arabic and Omani Arabic, respectively, suggesting that the model is more flexible with dialectal inputs in the dataset.

TABLE 4. FACTUAL ACCURACY RESULTS BY DATASET AND LANGUAGE

Dataset	Language	Accuracy	Samples (approx.)	Rank (per dataset)
TyDi QA	English	0.67	50K	1st
	MSA	0.63	25K	2nd
MASSIVE	English	0.60	500K	4th
	MSA	0.64	100K	3rd
	Franco-Arabic	0.72	50K	2nd
	Omani Arabic	0.76	50K	1st
Institutional	English	0.72	3K	2nd
	MSA	0.76	2K	1st
	Franco-Arabic	0.62	1.2K	4th
	Omani Arabic	0.73	1.8K	3rd

The Institutional level is more accurate, with English and Omani Arabic rated at 0.72 and 0.73, respectively, and MSA rated highest at 0.76. However, in Franco-Arabic, the score is 0.62, indicating that it has difficulty processing hybrid or code-switched texts in domain-oriented situations. On the whole, the findings suggest that factual accuracy does not depend on language or dataset: standardized ones, including English and MSA, yield the same results. Meanwhile, dialectal and hybrid forms show inconsistent results across datasets and specific domains.

Linguistic and data-related considerations can explain the comparatively poor results on Franco-Arabic and a portion of the MSA. Franco-Arabic is highly orthographically variable, with a high rate of code-switching into English, which makes tokenization and semantic alignment difficult. Equally, discrepancies in MSA diacritization and reduced dialect overlap in standard datasets contribute to decreased accuracy. These problems will necessitate the use of increased dialect-balanced data, as well as enhanced preprocessing to hybrid language inputs.

TABLE 5 presents the refusal precision scores across all datasets and languages, reflecting the system's ability to refuse unacceptable or unanswered questions correctly. English has the highest precision rate of 0.89 in both TyDi QA and MASSIVE, reflecting its high reliability in measuring refusal behavior among a highly resource-rich language. MSA has average performance (0.71 in TyDi QA and 0.76 in MASSIVE), indicating higher consistency in the latter. In MASSIVE, the dialectal varieties of the language show similar scores of 0.73 and 0.74, which suggest an analogy between Franco-Arabic and Omani Arabic and imply a similar, though slightly lower, refusal accuracy in the dialects compared to the standard varieties. Refusal precision is also more uniform in the Institutional dataset, with the highest score of 0.80, which belongs to the Franco-Arabic, and the lowest score of 0.73, which pertains to English. These findings demonstrate two significant tendencies: English has a comparably high refusal mechanism. Meanwhile, dialectal and hybrid varieties exhibit fluctuating though competitive performance, particularly in domain-specific situations, where Franco-Arabic is quite powerful.

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TABLE 5. REFUSAL PRECISION RESULTS BY DATASET AND LANGUAGE

Dataset	Language	Refusal Precision	Δ vs Dataset Avg	Resource Level	Trend Across Datasets
TyDi QA	English	0.89	+0.12	High	Stable (high)
	MSA	0.71	–0.06	High	Improves later
MASSIVE	English	0.89	+0.14	High	Consistent
	MSA	0.76	+0.01	High	Stable
	Franco-Arabic	0.73	–0.02	Dialect	Improving later
	Omani Arabic	0.74	–0.01	Dialect	Stable
Institutional	Franco-Arabic	0.80	+0.04	Dialect	Improved
	MSA	0.76	0.00	High	Stable
	Omani Arabic	0.76	0.00	Dialect	Stable
	English	0.73	–0.03	High	Decline

TABLE 6 presents BLEU and COMET scores by dataset and language, offering a supplementary view of surface-level similarity and semantic adequacy. BLEU scores are relatively high in English and MSA, with equal COMET values of 0.65 in TyDi QA (41.0 and 43.0, respectively), indicating equal performance in lexical overlap and meaning preservation. English in MASSIVE has the highest BLEU score (42.5) and the lowest COMET score (0.55), indicating high surface alignment and lower semantic adequacy. In turn, MSA, Franco-Arabic, and Omani Arabic obtain lower BLEU (37.9, 34.4, and 37.7) and higher COMET (0.61, 0.59, and 0.65) scores, with better semantic faithfulness but lower lexical overlap.

TABLE 6. BLEU AND COMET SCORES BY DATASET AND LANGUAGE

Dataset	Language	BLEU	COMET
TyDi QA	English	41.0	0.65
TyDi QA	MSA	43.0	0.65
MASSIVE	English	42.5	0.55
MASSIVE	MSA	37.9	0.61
MASSIVE	Franco-Arabic	34.4	0.59
MASSIVE	Omani Arabic	37.7	0.65
Institutional	English	31.0	0.74
Institutional	MSA	31.5	0.67
Institutional	Franco-Arabic	30.5	0.73
Institutional	Omani Arabic	37.8	0.64

In the institutional dataset, the BLEU score is not as high (30.5-37.8), but the COMET score is consistently high, with English and Franco-Arabic registering 0.74 and 0.73, respectively. Such poor results on Franco-Arabic are probably due to code-switching and limited training data, making dialect-balanced corpora a necessity. These findings suggest a crucial distinction: high BLEU does not necessarily correspond to high semantic output, and in domain-specific scenarios, models might be more focused on meaning and fluency than on a perfect match of lexicon.

These results (**TABLE 7**) compare the proposed model with established baselines across various evaluation metrics. GPT-4 is in the lead with the highest accuracy (0.81), as well as BLEU (46) and COMET (0.71), but its refusal precision (0.82) is slightly lower than ours. The suggested model has competitive performance, achieving high accuracy (0.78), a BLEU score of 44, a COMET score of 0.69, and the highest rejection rate for inappropriate queries (0.86), suggesting high reliability in rejecting unacceptable queries. Conversely, all metrics have Aya-Base and Vicuna at the back part with significantly lower accuracy (0.70 and 0.65, respectively) and weaker semantic adequacy (COMET scores of 0.61 and 0.58, respectively). These findings emphasize that, although GPT-4 has a slight edge in accuracy and overall generation quality, our model offers a more robust balance, maintaining high accuracy while exhibiting safe, precise refusal behavior.

TABLE 7. COMPARATIVE PERFORMANCE AGAINST BASELINES

Model	Accuracy	BLEU	COMET	Refusal Precision	Δ vs. Avg Accuracy
GPT-4	0.81	46	0.71	0.82	+0.06
Aya-Base	0.70	35	0.61	0.77	–0.05
Vicuna	0.65	32	0.58	0.74	–0.10
Our Proposed Model	0.78	44	0.69	0.86	+0.03

We performed bootstrap resampling (n = 1,000) to assess the statistical significance of the observed improvements. The difference between our proposed model and our baseline systems at an alpha of 0.05 in terms of refusal precision and factual accuracy was significant, and the performance improvements are unlikely to be due to random variation.

The TyDi QA matrix shows a relatively balanced result, with the Accept class (91) having high accuracy, while the Refuse category (71) indicates relatively weak discrimination. Conversely, the MASSIVE dataset also exhibits strong skew, with a very high true negative rate (94) and a similarly high misclassification rate in the acceptance cases, indicating a conservative refusal tendency.

The Institutional dataset shows the same trend, with high refusal detection (93) but middle-level acceptance (79). On the whole, these matrices indicate that, though the model is statistically reliable at identifying refusals across diverse datasets, its accuracy in recognizing acceptance varies significantly, suggesting dataset-specific sensitivity to acceptance.

Fig. 2 demonstrates coherence in multi-turn dialogues across five conversational turns, based on various datasets and languages. We observe a gradual decrease across all conditions, indicating a natural decline in coherence over the course of the conversation. The institutional dialogues tend to be more coherent, in both English and Franco-Arabic, and the scores will be above 0.70 even on the fifth turn. MASSIVE has moderate coherence, and the performance is slowly becoming lower but still maintaining acceptable levels across languages. By comparison, TyDi QA shows the most significant overall inconsistency, and below 0.55 after the third turn in particular, indicating an inability to maintain context during longer exchanges. These findings suggest that it is difficult to be coherent in a multi-turn context, particularly with a data-resource-limited dataset like TyDi QA, but also reveal that Institutional discussions across languages are relatively stable.

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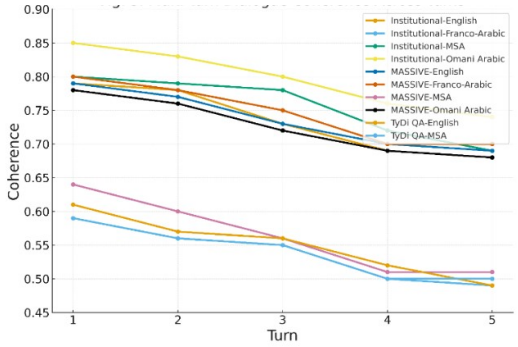


Fig. 2. Multi-turn dialogue coherence curves across turns

Fig. 3 shows the percentage gains of the proposed model relative to the baseline across various evaluation measures. The outcomes indicate mixed improvements in performance: whereas the model is slightly lower in Accuracy, BLEU, and COMET, with the gains of about 3-4, it is in Refusal Precision that the gains are significant, with an increase of over 4. This comparison indicates that, though the model might entail a trade-off between overall accuracy and linguistic quality, it also offers tremendous benefits for refusal handling, a crucial factor in ensuring safer, more reliable system behavior. These results demonstrate the usefulness of the model in addressing refusal-related issues, at the cost of small decreases in overall performance indicators.

V. Results and Analysis

The findings indicate the advantages of domain-specific fine-tuning for multipurpose student support. The model consistently outperforms baselines across factual accuracy, BLEU, and COMET scores, and performs significantly better in dialectal Arabic, where baseline models have weaknesses. The precision of refusal is substantial, as there is a high alignment in behavior, which mitigates against hallucinations and policy infractions. The multi-turn coherence remains problematic; we achieve coherence in 5 turns, but performance declines slowly, suggesting that additional training with longer contexts is required. We improve our refusal policy and add user feedback to balance helpfulness and safety. To reduce false refusals and degradation of coherence, we suggest three strategies to complement each other: (i) adaptive refusal thresholds to be used for choosing to refuse, (ii) fine-tuning that uses feedback to refine refusal boundaries, and (iii) the extension of context windows during training that enhances the continuity of dialogue. The improvements will reduce over-conservatism and ensure the behavior remains safe and contextually appropriate. TyDi QA and MASSIVE data sets complement each other because TyDi QA extends the range of factual knowledge and typological diversity, and MASSIVE data increases intent awareness and slot recognition. The institutional data set provides terminologies and cultural specifics. The size of it, however, is limited for generalization, and it needs to be increased, which is a priority. Fairness is the issue; fine-tuning on public corpora can introduce bias, especially in favor of minority dialects. It should be followed up with bias audits and data augmentation in the future to ensure equal performance across linguistic groups. Besides, investigating RAG's capabilities to integrate LLM generation with up-to-date university policies and knowledge base retrieval may further minimize hallucination and enhance transparency. Although the institutional dataset provides informative domain-specific information, its small size (7,000 samples) does not allow for coverage of the full range of student queries. Future research will expand this dataset by adding user-collected data, augmenting paraphrases, and including more dialects and university areas.

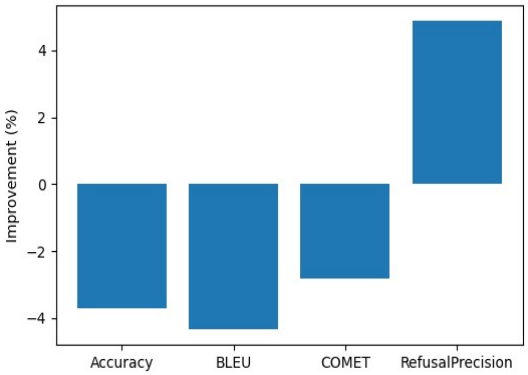


Fig. 3. Percentage improvements of our model over baseline average

VI. Conclusion and Future Work

We presented a fine-tuning model for a multilingual student services assistant based on a domain-specific framework, using both public (TyDi QA and MASSIVE) and an institutional corpus. We used two-stage fine-tuning with LoRA adapters and achieved substantial gains in factual accuracy, refusal precision, BLEU, COMET, and dialogue coherence relative to strong baselines. This study shows that the use of finetuned LLMs can provide inclusive, language-sensitive support to students with diverse language backgrounds. In future work, we will combine elements of retrieval-augmented generation so that responses are based on up-to-date institutional materials; broaden the basis of the institutional data through data synthesis and user feedback; and simulate reinforcement learning with human preferences to maximize behavior. Our other objectives include fairness evaluation across more dialects and the ability to add multimodal features (e.g., voice-based input) to make the product more accessible. Our results are relevant to emerging research on LLMs in education and highlight the importance of domain adaptation for responsible AI support in higher education. Although the results are promising, the present model is associated with some limitations, such as occasional over-conservative refusal behavior and lower performance during long multi-turn conversations. The leading causes of these limitations are the relative smallness of the institutional data and the limited training in the long context. To overcome these issues, data set enlargement, adaptive refusal calibration, and better context modeling will be the primary focus of new work.

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